

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026 — ANSWER KEY (Version 1)

Physics — Class 9 (Science Group) | Exam-13 | Total Marks: 31

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

SECTION A — MCQ Answers ($12 \times 1 = 12$ Marks)

1. Most reasonable estimate for the height of a classroom door → **B. 2 m** 1

A standard door is a little taller than a person, so roughly **2 m**. 0.2 m is the length of a pencil box, 20 m is a six-storey building and 200 m is taller than most towers. **Estimation** means choosing the size that makes physical sense.

2. Area of $4.5 \text{ cm} \times 2.11 \text{ cm}$ to correct significant figures → **C. 9.5 cm^2** 1

4.5 has **2** significant figures, 2.11 has **3**. In multiplication the answer carries the **smaller** number of significant figures, so the raw product 9.495 cm^2 is rounded to 2 s.f. → **9.5 cm^2** .

3. Value to report from five repeated readings → **C. 20.2 cm** 1

For repeated measurements the best value is the **average (mean)**: $(20.1 + 20.3 + 20.2 + 20.4 + 20.0) \div 5 = 101.0 \div 5 = \mathbf{20.2 \text{ cm}}$. Averaging cancels out **random errors**. 101.0 cm is the sum, not the average.

4. Universal speed limit → **C. $3 \times 10^8 \text{ m/s}$** 1

The **speed of light in vacuum** is about $3 \times 10^8 \text{ m/s}$ (300,000 km/s). No object or signal in the universe can travel faster than this.

5. Speedometer reading at one instant → **B. instantaneous speed** 1

Instantaneous speed is the speed at one particular moment of time. **Average speed** is total distance $\div$ total time for the whole journey, which a speedometer does not show.

6. Instrument that measures weight directly → **B. Force meter (spring balance)** 1

Weight is a **force**, so it is measured with a **force meter** (spring balance) calibrated in newtons. An electronic balance measures **mass** in kilograms, a metre rule measures length and a stopwatch measures time.

7. Non-contact force → **C. Magnetic force** 1

A **non-contact force** acts without the two bodies touching. Magnetic (also gravitational and electrostatic) forces act at a distance. Friction, tension and air resistance all need physical contact.

8. Second condition of equilibrium → **B. the sum of all torques is zero** 1

First condition: $\Sigma F = 0$ (no change in translational motion). **Second condition: $\Sigma \tau = 0$** (no change in rotational motion). For **complete** equilibrium both must hold.

9. The line along which a force acts → **B. line of action** 1

The **line of action** of a force is the straight line drawn along the direction in which the force acts. It matters when several forces act on one body, because it decides whether they turn the body or not.

10. Liquid pressure does NOT depend on → **C. the cross-sectional area of the container** 1

Liquid pressure is $P = \rho gh$. It depends only on **density, depth and g**. A narrow pipe and a wide tank filled to the same depth with the same liquid have exactly the same pressure at the bottom.

11. Material of solar photovoltaic cells → **B. silicon** 1

Solar (photovoltaic) cells are made of **semiconductor** materials, mainly **silicon**, which produce an electric current when sunlight falls on them. Silicon is also the second most abundant element in the Earth's crust.

12. Branch studying fundamental particles → B. Particle Physics 1

Particle Physics studies the fundamental building blocks of matter and the forces between them. Plasma Physics deals with ionised gases, Thermal Physics with heat, and Geophysics applies physics to the Earth.

SECTION B — Short Question Answers (3 × 3 = 9 Marks)

Q.2 (i) Resultant velocity of the swimmer; scalar or vector? 3

The two velocities are at **right angles**, so the resultant is found using Pythagoras' theorem:

$$v = \sqrt{v_N^2 + v_E^2} \quad 1$$

$$v = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25}$$

$$v = 5 \text{ m/s} \quad 1$$

Velocity is a **vector** quantity, because it needs **both a magnitude and a direction** to be described completely. (Speed, which has magnitude only, is a scalar.) 1

Q.2 (ii) Bus: acceleration in stage 1, and total distance over 35 s. 3

(a) Acceleration (first 10 s): $u = 0$, $v = 20 \text{ m/s}$, $t = 10 \text{ s}$

$$a = (v - u) / t = (20 - 0) / 10 = 2 \text{ m/s}^2 \quad 1$$

(b) Distance in stage 1 (accelerating):

$$s_1 = ut + \frac{1}{2}at^2 = (0)(10) + \frac{1}{2}(2)(10)^2 = \frac{1}{2} \times 2 \times 100 = 100 \text{ m} \quad 1$$

Distance in stage 2 (constant speed):

$$s_2 = v \times t = 20 \times 25 = 500 \text{ m}$$

$$\text{Total distance} = s_1 + s_2 = 100 + 500 = 600 \text{ m} \quad 1$$

Q.2 (iii) Gravitational field strength; weight on another planet; does mass change? 3 Definition:

Gravitational field strength at a point is the **gravitational force acting on unit mass** placed at that point.

Formula: $g = W / m$ (or $W = mg$) **SI unit:** N/kg 1

Weight on that planet: $m = 8 \text{ kg}$, $g = 3.7 \text{ N/kg}$

$$W = mg = 8 \times 3.7 = 29.6 \text{ N} \quad 1$$

Would the mass change? No. Mass is the **quantity of matter** in a body and stays the same everywhere in the universe. Only the **weight** changes, because weight depends on the value of g , which is different on each planet. 1

SECTION C — Long Question Answers (2 × 5 = 10 Marks)

Q.3 (i) Conditions of equilibrium + balancing the plank. 5**(a) The two conditions of equilibrium 2**

Condition	Statement	Mathematical form	Meaning
First condition	The vector sum of all the forces acting on the body is zero.	$\Sigma F = 0$ ($F_{\text{net}} = 0$)	No change in translational motion: the body stays at rest or moves with uniform velocity.
Second condition	The sum of all the torques acting on the body is zero.	$\Sigma \tau = 0$ ($\tau_{\text{net}} = 0$)	No change in rotational motion: the body does not start to spin.

For **complete equilibrium both** conditions must be satisfied at the same time.

(b) Finding the force F 3

Step 1 — distances from the pivot (the pivot is 1.2 m from the left end): **1**

Boy (left end, at 0 m) $\rightarrow 1.2 - 0 = 1.2$ m to the LEFT of pivot

Plank weight (centre, 1.8 m) $\rightarrow 1.8 - 1.2 = 0.6$ m to the RIGHT of pivot

Force F (right end, 3.6 m) $\rightarrow 3.6 - 1.2 = 2.4$ m to the RIGHT of pivot

The plank is **uniform**, so its whole weight acts at its **centre of gravity**, the mid-point at 1.8 m.

Step 2 — apply the principle of moments ($\Sigma \tau = 0$): **1**

Anticlockwise moment = Sum of clockwise moments

$$210 \times 1.2 = (90 \times 0.6) + (F \times 2.4)$$

$$252 = 54 + 2.4F$$

Step 3 — solve for F 1

$$2.4F = 252 - 54 = 198$$

$$F = 198 \div 2.4$$

F = 82.5 N, acting vertically downwards at the right-hand end.

Q.3 (ii) Power, crane numerical, energy conversion. 5**(a) Power 1**

Power is the **rate of doing work**, or the rate at which energy is converted from one form to another.

Formula: $P = W / t = E / t$ SI unit: **watt (W)**, where 1 W = 1 J/s. Power is a **scalar** quantity.

(b) The crane (m = 850 kg, h = 24 m, t = 40 s, g = 10 m/s²) 3

(i) Gain in potential energy 1

$$\text{P.E.} = mgh$$

$$\text{P.E.} = 850 \times 10 \times 24$$

$$\text{P.E.} = 204,000 \text{ J} = 2.04 \times 10^5 \text{ J} = 204 \text{ kJ}$$

(ii) Power output of the crane 2

The work done by the crane equals the potential energy gained, so $W = 204,000 \text{ J}$.

$$P = W / t$$

$$P = 204,000 \div 40$$

$$P = 5100 \text{ W}$$

$$\text{P} = 5.1 \text{ kW} \quad (\text{since } 1 \text{ kW} = 1000 \text{ W})$$

(c) Energy conversion in the motor 1

Electrical energy $\rightarrow$ mechanical energy, which is stored in the raised girder as **gravitational potential energy**.

— End of Answer Key —